

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a900436f9f2ced4a3.jsonl`  
- SHA-256: `2dfc4cfbc5c41d2c5a9e918bf2bbd7ee21afb7041ba604f5d670885e527eed37`  
- Source modified: `2026-05-31T04:22:01+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:21:06.040Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Pixtral-12B-2409 model weights are released under the Apache-2.0 license, making the weights commercially safe (not a restrictive custom/community/non-commercial license)."

****Source:**** <https://huggingface.co/mistralai/Pixtral-12B-2409> (primary)

****Supporting quote:**** "License: apache-2.0"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:21:08.481Z)

I'll verify this claim about Pixtral-12B-2409's license. Let me start by loading the search tools and checking the primary source.

3. assistant (2026-05-31T04:21:11.966Z)

I have the search tools. Let me verify the claim about Pixtral-12B-2409's license by checking the primary source and searching for any contradicting evidence. I'll run these checks in parallel.

4. user (2026-05-31T04:21:17.052Z)

License Information for Pixtral-12B-2409

The exact license listed for this model is **apache-2.0** (Apache License 2.0).

This is indicated in multiple places on the model card:

1. **License tag**: "License: apache-2.0" appears prominently in the model metadata section
2. **Key features section**: States "License: Apache 2.0"

There are no additional license statements or qualifications beyond the standard Apache 2.0 license designation.

5. user (2026-05-31T04:21:19.065Z)

Web search results for query: "Pixtral-12B-2409 license apache-2.0 Mistral weights"

Links: [{"title": "mistralai/Pixtral-12B-2409 · Hugging Face", "url": "https://huggingface.co/mistralai/Pixtral-12B-2409"}, {"title": "mistralai/Pixtral-12B-Base-2409 · Hugging Face", "url": "https://huggingface.co/mistralai/Pixtral-12B-Base-2409"}, {"title": "Pixtral-12B-2409 is now available on Amazon Bedrock Marketplace | Artificial Intelligence", "url": "https://aws.amazon.com/blogs/machine-learning/pixtral-12b-2409-is-now-available-on-amazon-bedrock-marketplace/"}, {"title": "Pixtral-12B: Pricing, Benchmarks & Performance", "url": "https://llm-stats.com/models/pixtral-12b-2409"}, {"title": "README.md · mistralai/Pixtral-12B-2409 at main", "url": "https://huggingface.co/mistralai/Pixtral-12B-2409/blob/main/README.md"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/html/2410.07073v2"}, {"title": "Pixtral 12B 2409 · Models · Dataloop", "url": "https://dataloop.ai/library/model/mistralai_pixtral-12b-2409/"}, {"title": "Pixtral 12B 2409 by Mistral AI on Vercel AI Gateway, Specs, Pricing & API", "url": "https://vercel.com/ai-gateway/models/pixtral-12b"}, {"title": "Pixtral-12B-2409 is now available on Amazon Bedrock Marketplace - HKU SPACE AI Hub", "url": "https://aihub.hkuspace.hku.hk/2025/03/04/pixtral-12b-2409-is-now-available-on-amazon-bedrock-marketplace/"}, {"title": "[Deprecated] Pixtral 12B | Mistral AI", "url": "https://mistral.ai/news/pixtral-12b/"}]

Based on the search results, here's what I found about Pixtral-12B-2409:

License

Pixtral-12B-2409 is licensed under Apache 2.0. Similar to other Mistral models such as Mistral 7B, Mixtral 8x7B, Mixtral 8x22B, and Mistral Nemo 12B, Pixtral 12B is released under the commercially permissive Apache 2.0.

Model Architecture and Weights

The Pixtral-12B-2409 is a Multimodal Model of 12B parameters plus a 400M parameter vision encoder. It was released with open weights under an Apache 2.0 license.

Pixtral is an instruction tuned model which is pretrained on large scale interleaved image and text documents, and hence is capable of multi-turn, multi-image conversation.

The model weights are available from Mistral AI on Hugging Face (mistralai/Pixtral-12B-2409), where they can be downloaded and used under the Apache 2.0 license.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:21:22.091Z)

Both the primary source and search results confirm Apache-2.0. Let me cross-check with Mistral's official announcement and corroborate the weights-license distinction (the claim specifically asserts the WEIGHTS, not just the repo, are Apache-2.0).

7. user (2026-05-31T04:21:26.050Z)

Pixtral 12B License

According to the webpage, Mistral states that Pixtral 12B uses an **Apache 2.0 license**.

The verbatim statement from the page reads:

```
> "License: Apache 2.0"
```

This is a permissive open-source license that allows for commercial and non-commercial use with minimal restrictions.

8. user (2026-05-31T04:21:30.472Z)

Web search results for query: "Pixtral 12B Mistral Research License non-commercial OR restricted weights"

Links: [{"title": "Pixtral 12B: Specs, Benchmarks & How to Deploy Mistral's Vision Model (2026)", "url": "https://ucstrategies.com/news/pixtral-12b-specs-benchmarks-how-to-deploy-mistrals-vision-model-2026/"}, {"title": "mistralai/Pixtral-12B-2409 · Hugging Face", "url": "https://huggingface.co/mistralai/Pixtral-12B-2409"}, {"title": "[Deprecated] Pixtral 12B | Mistral AI", "url": "https://mistral.ai/news/pixtral-12b/"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/html/2410.07073v2"}, {"title": "Mistral Unveils Multimodal Pixtral 12B | ml-news", "url": "https://wandb.ai/byyoung3/ml-news/reports/Mistral-Unveils-Multimodal-Pixtral-12B--Vmlldzo5MzMNzUx"}, {"title": "Pixtral-12B: Pricing, Benchmarks & Performance", "url": "https://llm-stats.com/models/pixtral-12b-2409"}, {"title": "Mistral just released Pixtral-12B their first multi model | by Mirza Samad | Major Digest | Medium", "url": "https://medium.com/majordigest/mistral-just-released-pixtral-12b-their-first-multi-model-4962fa9c6edc"}, {"title": "Mistral AI Unveils Pixtral 12B, a Multimodal AI Model for Text and Image Processing", "url": "https://www.maginitive.com/article/mistral-ai-unveils-pixtral-12b-a-multimodal-ai-model-for-text-and-image-processing/"}, {"title": "Pixtral 12B ? Try & Compare | Roboflow Playground", "url": "https://playground.roboflow.com/models/mistral/pixtral-12b"}, {"title": "mistral-community/pixtral-12b-240910 · License?", "url": "https://huggingface.co/mistral-community/pixtral-12b-240910/discussions/7"}]

Based on the search results, I can provide you with information about Pixtral 12B's licensing:

Pixtral 12B is released under the Apache 2.0 license. The Apache 2.0 license has no export restrictions, unlike some Chinese models that face US trade controls.

Regarding your specific query about non-commercial or restricted weights: The search results do not show that Pixtral 12B has a "Mistral Research License" with non-commercial restrictions. Released in September 2024 under Apache 2.0 license, the model is free to download and use locally (open weights under Apache 2.0).

One source mentions that Mistral has emphasized making Pixtral available to researchers and academics, allowing free access to the model via platforms like GitHub and Hugging Face, but commercial users need a paid license. However, this appears to describe an older or different licensing arrangement, as the official current license is Apache 2.0, which is free for everything.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:21:33.923Z)

I have strong, consistent corroboration. Let me do one final adversarial check by looking directly at the raw README metadata on Hugging Face to confirm the `license: apache-2.0` tag is in the model card's YAML frontmatter (the authoritative declaration), and address the one ambiguous note about "commercial users need a paid license."

10. user (2026-05-31T04:21:36.437Z)

License Field from YAML Frontmatter

The exact value set for the license field is:

```
...  
apache-2.0  
...
```

11. assistant (2026-05-31T04:21:48.828Z)

The evidence is conclusive and consistent across all sources. Let me evaluate against the adversarial checklist:

1. **Is the claim supported by the quote?** Yes. The quote "License: apache-2.0" directly supports the claim. The claim is about the WEIGHTS being Apache-2.0, and the authoritative YAML frontmatter `license: apache-2.0` on the official `mistralai/Pixtral-12B-2409` repo (where the weights themselves are hosted) declares exactly this. The license tag on a HF weights repo governs the weights.
2. **Contradicting evidence?** None credible. I deliberately searched for "Mistral Research License / non-commercial / restricted weights" to refute it. The only hint of restriction was a vague secondary-source sentence ("commercial users need a paid license") that the search engine itself flagged as describing "an older or different licensing arrangement" and explicitly corrected ? this conflates Pixtral with Mistral's *other* models (e.g., Pixtral Large, Mistral Large) which DO use the non-commercial MRL. The 12B (2409) is consistently Apache-2.0.
3. **Source quality?** Strong. Primary source = official Mistral HF repo YAML frontmatter (the authoritative machine-readable license declaration), plus Mistral's own announcement, plus AWS Bedrock. This is a verifiable, non-extraordinary claim well-matched by primary sources.
4. **Outdated?** No. The model is from Sept 2024; the Apache-2.0 designation is stable and current as of 2026. Note: Mistral's *newer/larger* vision models shifted to MRL, but that does not retroactively change the 12B-2409 license.
5. **Marketing/cherry-picked?** No. A license tag in repo metadata is a factual, contractual declaration, not marketing.

One important nuance worth flagging: the claim is narrowly scoped to **Pixtral-12B-2409** specifically ? which is correct. This must NOT be generalized to "all Pixtral/Mistral vision models," since Pixtral Large and several other Mistral models use the restrictive MRL (non-commercial). But the claim as written is correct.

Relevant file in repo: `C:\Users\avidu\Projects\muneem\docs\DEPENDENCIES.md` (license-tracking is an active concern per the project's commercial-license-safety memory).

12. user (2026-05-31T04:21:57.252Z)

Structured output provided successfully

13. assistant (2026-05-31T04:22:01.433Z)

Claim verified as NOT refuted. The Apache-2.0 license for Pixtral-12B-2409 weights is confirmed by the authoritative YAML frontmatter in the official HuggingFace weights repo, corroborated by Mistral's own announcement and AWS Bedrock documentation. I flagged the one important scoping nuance (do not generalize to Pixtral Large / other Mistral VLMS, which use the restrictive non-commercial MRL) ? but as narrowly written, the claim is accurate and well-sourced.